

MPCA Module 1 — 2 Mark Questions & Answers

Q1. Define Computer Architecture and Computer Organization.

- **Computer Architecture:** Refers to the functional attributes of a computer system as seen by the programmer (e.g., instruction set, data types, addressing modes).
 - **Computer Organization:** Refers to the operational units and their interconnections that realize the architectural specifications (e.g., clock frequency, control signals, memory technology).
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Q2. What is the Von Neumann bottleneck?

The **Von Neumann Bottleneck** is a speed limitation caused by sharing a single physical bus for both instruction fetch and data transfer. Since the CPU cannot fetch an instruction and read/write data simultaneously, the throughput of the system is constrained by the speed of the shared bus.

Q3. Define CPI and IC with formulas.

- **CPI (Cycles Per Instruction):** The average number of clock cycles required to execute a single instruction.
 - $$\text{CPI} = \frac{\text{Total CPU Clock Cycles}}{\text{Instruction Count}}$$
 - **IC (Instruction Count):** The total number of machine instructions executed by a program during runtime.
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Q4. What is the stored program concept?

The **Stored Program Concept** (proposed by Von Neumann) states that both program instructions and data are stored together in the same main memory (RAM) in binary format. This allows the computer to fetch, decode, and execute instructions sequentially without requiring physical hardware rewiring.

Q5. State two differences between Von Neumann and Harvard architecture.

Feature	Von Neumann Architecture	Harvard Architecture
Bus System	Single shared bus for instructions and data	Separate buses for instructions and data
Memory	Shared memory space for code and data	Physically separate memory units for code and data

Q6. What are the two units of the 8086 microprocessor?

1. **Bus Interface Unit (BIU):** Handles all bus operations, address calculation, memory fetching, and instruction queuing.
 2. **Execution Unit (EU):** Decodes and executes instructions fetched by the BIU using the ALU and registers.
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Q7. Define MIPS and MFLOPS.

- **MIPS (Millions of Instructions Per Second):** A measure of processor execution speed.
- $$\text{MIPS} = \frac{\text{Instruction Count}}{(\text{Execution Time} \times 10^6)}$$

- **MFLOPS (Millions of Floating-Point Operations Per Second):** A measure of processor floating-point calculation performance.
 - $\text{MFLOPS} = \text{Floating-Point Operations} / (\text{Execution Time} \times 10^6)$
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Q8. What is the size of the pre-fetch queue in 8086?

The 8086 microprocessor features a **6-byte** pre-fetch instruction queue located inside the Bus Interface Unit (BIU). It prefetches up to 6 bytes of instruction code from memory to enable instruction pipelining.

Q9. State the syntax and one example of immediate addressing mode.

In **Immediate Addressing Mode**, the operand data is specified directly within the instruction itself.

* **Syntax:** `MOV destination, immediate_value`

* **Example:** `MOV AX, 1234H` (Loads the 16-bit hexadecimal constant `1234H` directly into register `AX`).

Q10. Differentiate between minimum mode and maximum mode of 8086.

- **Minimum Mode (MN/MX' = 5V):** Configured for single-processor systems; the 8086 directly generates all bus control signals itself.
 - **Maximum Mode (MN/MX' = 0V):** Configured for multi-processor / co-processor systems (e.g., 8086 + 8087); control signals are generated by an external 8288 Bus Controller.
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Q11. What is the function of BIU in 8086?

The **Bus Interface Unit (BIU)** manages all external data, address, and control bus operations. Its primary functions are:

1. Fetching instruction bytes from memory into the 6-byte queue.
 2. Calculating 20-bit physical memory addresses ($\text{Physical Address} = \text{Segment} \times 16 + \text{Offset}$).
 3. Reading and writing data operands to/from memory and I/O ports.
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Q12. Define latency and throughput.

- **Latency (Execution Time):** The total time required to complete a single task from start to finish (measured in seconds or milliseconds).
 - **Throughput:** The total amount of work or number of tasks completed per unit of time (e.g., instructions per second).
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